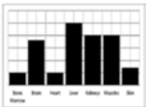
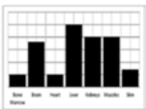
## 1.1.e. Effects of exercise on body systems

Learners will develop their knowledge and understanding of the short and long-term effects of exercise on muscles and bones, the heart and the respiratory system. They will be able to apply understanding of these effects to examples from a range of physical activities and sports.

Learners will be able to collect and use data in this section related to both short-term and long-term effects of exercise.

| Topic area | Learners must: |
|---|---|
| Short-term effects of exercise<br><br> | • understand the short-term effects of exercise on:<br>○ muscle temperature<br>○ heart rate, stroke volume, cardiac output<br>○ redistribution of blood flow during exercise<br>○ respiratory rate, tidal volume, minute ventilation<br>○ oxygen to the working muscles<br>○ lactic acid production<br>• be able to apply the effects to examples from physical activity/sport<br>• be able to collect and use data relating to short-term effects of exercise. |
| Long-term (training) effects of exercise<br><br> | • understand the long-term effects of exercise on:<br>○ bone density<br>○ hypertrophy of muscle<br>○ muscular strength<br>○ muscular endurance<br>○ resistance to fatigue<br>○ hypertrophy of the heart<br>○ resting heart rate and resting stroke volume<br>○ cardiac output<br>○ rate of recovery<br>○ aerobic capacity<br>○ respiratory muscles<br>○ tidal volume and minute volume during exercise<br>○ capilliarisation<br>• be able to apply the effects to examples from physical activity/sport<br>• be able to collect and use data relating to long-term effects of exercise. |